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Feasibility of Using Mixed-Oxide Fuel for a Pressurized Water Reactor Instead of Traditional UO_2 Fuel Material

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Abstract — This study examines the feasibility of utilizing mixed-oxide fuel $[(\text{U}_{0.9}, \text{rgPu}_{0.1}) \text{O}_2]$ instead of traditional UO_2 in nuclear reactors. The utilization of $(\text{U}_{0.9}, \text{rgPu}_{0.1}) \text{O}_2$ is particularly significant as it represents an effective approach to nuclear fuel recycling by combining reactor-grade plutonium extracted from partially used nuclear fuel and depleted uranium obtained through the enrichment process. The fundamental neutronic characteristics, such as the radial power distribution, were investigated using the MCNPX 2.7 algorithm to identify the specific channel for subsequent thermal-hydraulic (TH) analysis. The TH analysis was conducted using COMSOL-Multiphysics, allowing for the estimation of the fuel rod's axial and radial temperature profiles, as well as the determination of the departure from the nucleate boiling ratio. Furthermore, the coupling between heat transfer and solid structure (SS) was achieved using the Multiphysics tool in COMSOL-Multiphysics. This coupling facilitated the simulation of key SS parameters, including von Mises stress, volumetric strain, and displacement, while considering the influence of heat transfer. The results demonstrate significant improvements and enhanced safety margins when utilizing $(\text{U}_{0.9}, \text{rgPu}_{0.1}) \text{O}_2$.

Keywords — Mixed oxide, MCNPX, neutronic analysis, COMSOL-Multiphysics, thermal hydraulics.

Note — Some figures may be in color only in the electronic version.

I. INTRODUCTION

Nuclear power plays a crucial role in meeting the world's energy needs, offering a reliable and low-carbon source of electricity. However, safety concerns and the desire for improved reactor performance have driven research into accident-tolerant fuels (ATFs) as potential alternatives to conventional nuclear fuel (UO_2). These fuels aim to enhance safety, increase fuel efficiency, and extend fuel lifetimes.^[1] Conventional nuclear fuel, typically based on uranium dioxide (UO_2), has been used extensively in pressurized water reactors (PWRs) because of its favorable properties. However, during severe

accidents, such as loss-of-coolant incidents or extended station blackouts, fuel integrity may be compromised, potentially leading to fuel failure and the release of radioactive materials.^[2,3] To address these challenges, advanced fuel concepts like ATFs have emerged.

Nuclear waste poses significant hazards because of its radioactivity and long half-life. The accumulation of nuclear waste requires safe disposal methods to prevent environmental contamination and potential health risks. However, a proposed solution to address these hazards is the feasibility of reusing nuclear waste as an ATF. By repurposing the waste material, such as depleted uranium and partially used nuclear fuel, as fuel for reactors, we can potentially mitigate the risks associated with long-term storage and disposal. This approach aims to

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maximize the energy extraction from nuclear fuel while minimizing the amount of waste that requires long-term storage.

By using nuclear waste as an ATF, we can both reduce the volume of waste and harness its energy potential, contributing to a more sustainable and efficient use of nuclear resources. However, the implementation of this solution requires careful evaluation of technical, safety, and regulatory aspects to ensure the safe handling, performance, and environmental impact of using nuclear waste in this manner. Mixed-oxide (MOX) fuel consists of a blend of reactor-grade plutonium (rgPu) and depleted uranium, offering several advantages as an ATF. The inclusion of rgPU enhances the thermal properties and neutronic behavior of the fuel, potentially improving fuel performance and safety characteristics.^[4]

Various codes are utilized for neutronic analysis, such as MCNP codes,^[5] SERPENT, and OPENMC, that use the stochastic approach to solve the neutron transport equation. For the thermal-hydraulic (TH) performance, the rod-centered subchannel model, the most relevant, is solved using either an analytical or numerical approach. This subchannel model aids in the computation of various TH parameters, including the coolant pressure drop, coolant enthalpy change, actual heat flux ($\dot{q}_{act}$), temperature profiles in both the radial and axial directions, and the departure from nucleate boiling ratio (DNBR). To simulate the subchannel model numerically, the subsequent partial differential equations (PDEs) were solved together [heat generation, momentum (Navier-Stokes), and continuity equations]. Several computational fluid dynamics tools are employed to simulate this coupling numerically, such as COMSOL-Multiphysics and ANSYS Fluent.^[6,7]

In this study, we investigate the feasibility of using MOX fuel in a PWR as an ATF through highlighting its benefits and possible drawbacks from the perspectives of economics, neutronics, TH, and solid structure (SS). Computational simulations were done by using MCNPX and COMSOL-Multiphysics to analyze its neutronic and thermal behavior, respectively.

II. METHODOLOGY

II.A. Neutronic Analysis

The neutron transport equation is an integral-differential equation that describes the behavior of neutrons as they move through a medium. In MCNPX, the neutron transport equation is solved using the Monte Carlo method, which involves simulating the behavior

of individual neutrons. The governing equation for neutron transport can be written as follows:

$$\begin{aligned} \nabla \cdot (\Omega \nabla \Phi(r, \Omega, E)) + \sigma_t(r, E) \Phi(r, \Omega, E) \\ = \int dE' \int d\Omega' \sigma_s(r, E', \Omega' \rightarrow E, \Omega) \Phi(r, \Omega', E') \\ + Q(r, \Omega, E) \end{aligned} \quad (1)$$

where

$\Phi(r, \Omega, E)$ = neutron angular flux at position r , moving in the direction Ω and with energy E

$\sigma_t(r, E)$ = total macroscopic cross section at position r and energy E , which accounts for all neutron interactions

$d\Omega' \sigma_s(r, E', \Omega' \rightarrow E, \Omega)$ = scattering macroscopic cross section representing the probability of a neutron with energy E' and direction Ω' scattering to energy E and direction Ω

$Q(r, \Omega, E)$ = neutron source term that accounts for the external neutron sources in the system.

In the Monte Carlo method, individual neutrons are tracked through the system and their interactions are simulated based on probabilities derived from cross-section data. By simulating the behavior of a large number of neutrons and statistically analyzing the results, MCNPX provides a solution to the neutron transport equation, allowing for the calculation of various quantities, such as neutron flux, reaction rates, and radiation dose distributions. The MCNPX simulations were done on one of the central core fuel assemblies' cell calculations by MCNPX 2.7 employing libraries (ENDF/B-V-II).^[5] Using 30 000 histories, 200 active cycles, and 50 cycles that were skipped, the MCNPX criticality calculations were carried out to achieved standard deviation of about 10 pcm.

II.B. TH and Solid Mechanics Analysis

The coupling between heat transfer and fluid flow is a fundamental aspect of many engineering applications. Understanding and accurately modeling the interactions between heat transfer and fluid dynamics is crucial for ensuring safe operation and improving overall efficiency.

COMSOL-Multiphysics, a finite element analysis software, provides advanced capabilities for simulating and analyzing such coupled phenomena. By integrating the governing equations for heat transfer and fluid flow, which are described in the following, along with appropriate boundary conditions, COMSOL-Multiphysics allows for comprehensive investigations into complex thermal-fluid systems.

In this paper, we explore the coupling between heat transfer and fluid flow using COMSOL-Multiphysics, focusing on the utilization of the finite element method (FEM) and the inclusion of a turbulence k-epsilon model to capture turbulent effects. Through the integration of these techniques, we aim to gain a deeper understanding of the coupled physics and to provide valuable insights for engineering applications requiring accurate thermal-fluid simulations.

The governing equations are

Conservation of mass:

$$\nabla \cdot (\rho u) = 0 \quad , \quad (2)$$

Conservation of momentum:

$$\rho \left(\frac{\partial u}{\partial t} + u \cdot \nabla u \right) = -\nabla p + \mu (\nabla \cdot \nabla u) + \rho g + F_{turb} \quad , \quad (3)$$

Conservation of energy:

$$\rho C_P \left(\frac{\partial T}{\partial t} u \cdot \nabla T \right) = \nabla \cdot (k \nabla T) + Q \quad , \quad (4)$$

where

ρ = density

u = velocity vector

p = pressure

μ = dynamic viscosity

g = acceleration because of gravity

F_{turb} = turbulent fluid force

T = temperature

k = thermal conductivity

C_P = specific heat capacity

Q = heat source term.

The k-epsilon turbulence model is used to capture the effects of turbulence in the flow field. It provides additional equations to solve for the turbulence kinetic energy

(k) and its dissipation rate (epsilon). The transport equations for k and epsilon can be written as

$$\frac{\partial \rho K}{\partial t} + \nabla \cdot (\rho K u) = \nabla \cdot (\mu_t \nabla K) + P_K - \varepsilon \quad (5)$$

$$\begin{aligned} \frac{\partial \rho \varepsilon}{\partial t} + \nabla \cdot (\rho \varepsilon u) = & \nabla \cdot (\mu_t \nabla \varepsilon) + C_{\varepsilon 1} \frac{\varepsilon}{K} (P_K - \varepsilon) \\ & - C_{\varepsilon 2} \frac{\varepsilon^2}{K} \quad , \end{aligned} \quad (6)$$

where

μ_t = turbulent viscosity

P_K = production of K because of mean velocity gradients

ε = dissipation rate of K

$C_{\varepsilon 1}, C_{\varepsilon 2}$ = empirical constants.

To simulate the SS parameters, three sets of equations from elasticity theory are employed. First, there are the differential equations of equilibrium, which describe the forces acting on the model, including normal forces generating normal stress σ , shear forces producing shear stress τ , and body forces due to gravity, as illustrated in Eq. (7):

$$\sum F = F_{normal} + F_{shear} + F_{body} = 0 \quad . \quad (7)$$

Second, the strain/displacement relationships are utilized to derive strain from displacement, as indicated in Eq. (8):

$$\{\varepsilon\} = [B] \times \{d\} \quad . \quad (8)$$

In this particular case study, thermal stress is predominant, making displacement highly dependent on the thermal expansion coefficients of the fuel and clad materials. Once the strain is determined, the third set of equations (stress/strain relationships) is applied to calculate the stress acting on the material, as demonstrated in Eq. (9)^[8]:

$$\{\sigma\} = [D] \times \{\varepsilon\} \quad , \quad (9)$$

where $\{\sigma\}$ is the stress vector

$$\{\sigma\} = \begin{bmatrix} \sigma_x \\ \sigma_y \\ \sigma_z \\ \tau_{xy} \\ \tau_{yz} \\ \tau_{zx} \end{bmatrix}, \{\epsilon\} = \begin{bmatrix} \epsilon_x \\ \epsilon_y \\ \epsilon_z \\ \gamma_{xy} \\ \gamma_{yz} \\ \gamma_{zx} \end{bmatrix}, \{\epsilon\} \text{ is the strain vector, and } [D] \text{ is the constitutive matrix}$$

$$[D] = \frac{E}{(1+\nu)(1-2\nu)} \begin{bmatrix} 1-\nu & \nu & \nu & 0 & 0 & 0 \\ \nu & 1-\nu & \nu & 0 & 0 & 0 \\ \nu & \nu & 1-\nu & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1-2\nu}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1-2\nu}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1-2\nu}{2} \end{bmatrix}.$$

In this study, the TH analysis was performed using COMSOL-Multiphysics employing the FEM, which is used to discretize the governing equations and solve them numerically. Then through the meshing tool, the computational domain is divided into a mesh of finite elements, and within each element, approximate solutions are obtained using shape functions. The governing equations are then converted into a system of algebraic equations by integrating them over each element and applying the weak form. The resulting system of equations is solved iteratively using numerical techniques (e.g., Newton's method) until a converged solution is obtained.

To solve the coupled problem of heat transfer and fluid dynamics, the governing equations for momentum, energy, and turbulence are solved simultaneously. The equations are coupled through the velocity, pressure, and temperature fields. The solution procedure involves iteratively solving the equations for each time step or until a steady-state solution is reached. The solution process involves updating the velocity, pressure, temperature, turbulent quantities, and other relevant variables until convergence is achieved.

In our earlier study, we exhibited the design specifications and the schematic view for the standard AP-1000 fuel assembly.^[9,10] COMSOL-Multiphysics computer software was used to distinguish between the TH performance of both the traditional fuel material of UO₂ and the proposed material of (U_{0.9}, GdPu_{0.1}) O₂. Because of its control over the precision and convergence of the solution, the meshing procedure is a crucial stage in the numerical solution. Therefore, in order to obtain the accurate result to any numerical problem, a mesh-independent solution must be done to examine how the primary TH or SS parameters fluctuate as the mesh size changes. With decreasing mesh size under the condition of a good mesh quality, the solution approaches the mesh-independent solution.

Mesh quality is a measure of a computational mesh's ability to accurately capture the geometrical characteristics of the physical domain being simulated. Typically, it is measured using a metric that ranges from 0 to 1, with 0 denoting extremely low mesh quality and 1 denoting exceptional mesh quality. The mesh that achieved the mesh-independent solution has the following characteristics: total number of elements in this particular mesh configuration is 784 102, with an average element quality of 0.7816 and a minimum element quality of 0.2001.

III. RESULTS AND DISCUSSION

III.A. MCNPX Code Verification

This section aims to verify the obtained results by comparing them with the design control document (DCD). The AP-1000 DCD is an extensive technical dossier furnished by Westinghouse Electric Company to both regulatory authorities and potential customers interested in the AP-1000 nuclear reactor design. This document functions as a detailed roadmap for shaping the reactor's design, building process, and operational guidelines. It holds paramount importance within the regulatory clearance procedure. The DCD is structured into several essential segments, encompassing the Reactor Design Description, Safety Analysis, Operational Procedures, Environmental Impact Assessment, Licensing Information, Quality Assurance, and Testing and Inspection Protocols.^[11] For the criticality results, specifically for the maximum K_{inf} that occurs at cold shutdown condition, the obtained value was 1.32878, while in the DCD it was 1.328.

Both the moderator temperature coefficient (MTC) and the fuel temperature coefficient (FTC) have been computed and compared to the DCD. The major objective of the reactivity calculation in nuclear reactors is to determine and quantify the changes in neutron multiplication and power levels resulting from the variations in the fuel and moderator properties. Reactivity calculations are essential for ensuring the safe and efficient operation of nuclear reactors. By accurately estimating the reactivity, operators can control the power level, maintain criticality, and optimize reactor performance. Equation (10) is used to calculate the reactor reactivity:

$$\rho = \frac{K_{inf} - 1}{K_{inf}}. \quad (10)$$

By differentiating Eq. (10) with regard to temperature, Eq. (11) is obtained to calculate both the FTC and MTC^[12]:

$$\alpha(T) = \frac{1}{K_{inf}^2} \times \frac{dK_{inf}}{dT} \quad . \tag{11}$$

For fuel burnup results, both the fuel cycle length (FCL) and maximum fuel burnup have been compared with the DCD. The obtained FCL (full-cycle length) was 480 days, while it was from 360 to 540 days in the DCD. Also, the maximum fuel burnup value herein was 54.63 (GWd/tonne U), while in the DCD it was equal to 60 to 62 (GWd/tonne U). Finally, the comparison was done on the power distribution results, specifically on the power peaking factor (PPF). The PPF is determined by dividing the power generated on each fuel rod, which can be estimated using MCNPX by average power produced per fuel rod. The maximum PPF obtained herein by using MCNPX 2.7 was 1.357, which is very close to its corresponding value in the DCD, which was 1.396.

According to Table I, the obtained results are within the best-estimate values based on the DCD. The slight deviation in the results is due to the difference in the calculation method, where in the DCD, most of the results are based on mainly experimental works.

III.B. Neutronic Analysis Results

III.B.1. Fuel Depletion

The fuel burnup card is a prominent feature that has been added to the MCNPX version, providing convenient access to information about the examined fuel, including the FCL, maximum fuel burnup, and production of actinide and nonactinide nuclides. An analysis of Figs. 1a and 1b reveals the variation of K_{inf} (infinite multiplication factor) with full effective power days (FEPDs) and fuel burnup, respectively. As both fuel types (MOX and UO₂)

lack burnable absorbers to regulate fuel depletion and reactor reactivity, their reactivity experiences a pronounced and approximately linear decline. It is evident from Fig. 1a that the MOX fuel material has a threefold higher FCL compared to traditional fuel material. This can be attributed to the larger average neutron yield v_{avg} of (U_{0.9}, rgPu_{0.1}) O₂, as depicted in Fig. 1c, which enhances the possibility of fast fission reactions involving ²³⁹Pu.

Figures 2a and 2b demonstrate a higher consumption of both ²³⁵U and ²³⁸U in the MOX fuel material compared to the traditional one, as the higher v_{avg} in the MOX fuel material increases the likelihood of fission reactions, which increases the consumption of both ²³⁵U and ²³⁸U. Regarding spent fuel in the MOX fuel, accurate storage calculations (shielding properties and time of storage) are required because of its high production of radioactive materials. Therefore, investigating the nuclide inventory formed during reactor operation in the case of using MOX fuel becomes crucial.

Figure 3a displays the inventory of minor actinides in both the MOX and traditional fuel (UO₂) materials, with particular emphasis on their lengthy half-lives, such as ²⁴¹Am and ²³⁷Np. The production of nonactinide nuclides, specifically ¹³⁵Xe and ¹⁴⁹Sm, is examined in Figs. 3b and 3c because of their significant thermal neutron capture, which impacts reactor reactivity. Notably, the MOX fuel material exhibits a larger production of nonactinide and minor actinide nuclides compared to the UO₂, which represents a drawback when utilizing (U_{0.9}, rgPu_{0.1}) O₂ as an ATF.

In Fig. 3b, it is evident that the generation of ¹³⁵Xe experiences a surge at the onset of the fuel cycle owing to the heightened fission rate at the beginning of the cycle. Subsequently, the production rate of ¹³⁵Xe from fission reactions aligns closely with the decay rate of ¹³⁵Xe,

TABLE I
Obtained Neutronic Results That Are Utilized in Verifying the Applied MCNPX Code

Point of View	Parameter	Value	DCD Best Estimate ^[11]	DCD Design Limits ^[11]
Criticality	Maximum K_{inf}	1.32878	1.328	
Reactivity	FTC (pcm)/°C	−2.158	−3.8 to −2.3	−6.3 to −1.8
	MTC (pcm)/°C	−12.4	0 to −63	0 to −72
Fuel burnup	FCL (days)	480	360 to 540	
Power	Maximum fuel burnup (GWd/tonne U)	54.63	60 to 62	
	Maximum PPF	1.357	1.396	

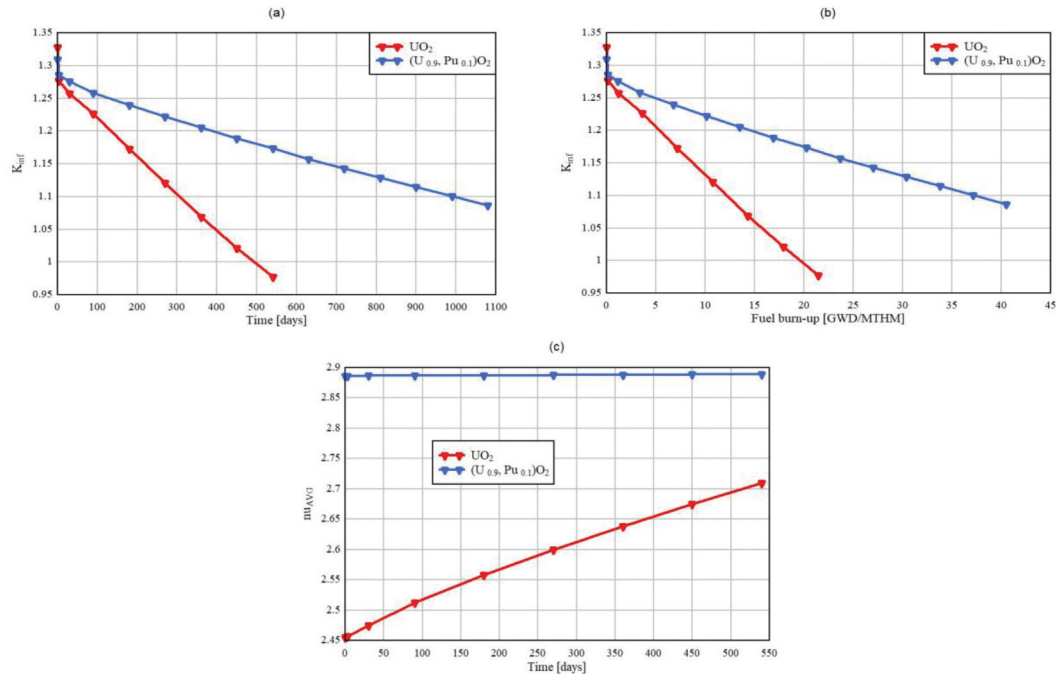


Fig. 1. Fuel burnout outcomes for (a) FCL, (b) fuel burnup, and (c) v_{avg} .

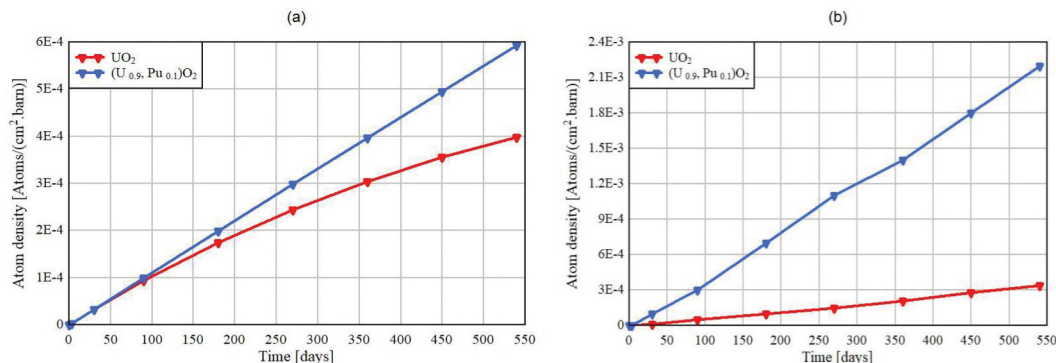


Fig. 2. Consumption for (a) ^{235}U and (b) ^{238}U .

given its exceedingly brief half-life of approximately 9.2 h. This results in a swift decline in its radioactivity over just a few days. This dynamic constitutes the primary factor influencing the behavior of Xe production over time of an initial sharp increase at the onset of the fuel cycle, followed by production levels that nearly mirror depletion rates.

Moving on to the production of Sm, as depicted in Fig. 3b over the FEPDs, a similar trend is observed for ^{149}Sm in UO_2 like that of ^{135}Xe , as the production rate of ^{149}Sm is very small and compatible with its decay. However, in the case of MOX fuel, the production rate of ^{149}Sm surpasses the rate of consumption. It is worth noting that ^{149}Sm exhibits a notably stable

nature, boasting a half-life of approximately 2.8×10^{11} years, resulting in an extremely small decay rate. This, in turn, contributes to an augmentation in Sm production throughout the fuel cycle duration.

III.B.2. Flux and Power Distribution Results

To identify the hot channels within the fuel assemblies, simulations were conducted to obtain the radial PPF profiles for both the MOX and UO_2 configurations. As depicted in Figs. 4a and 4b, the maximum PPF for the UO_2 fuel assembly was found to be 1.357, whereas for the MOX fuel assembly it was

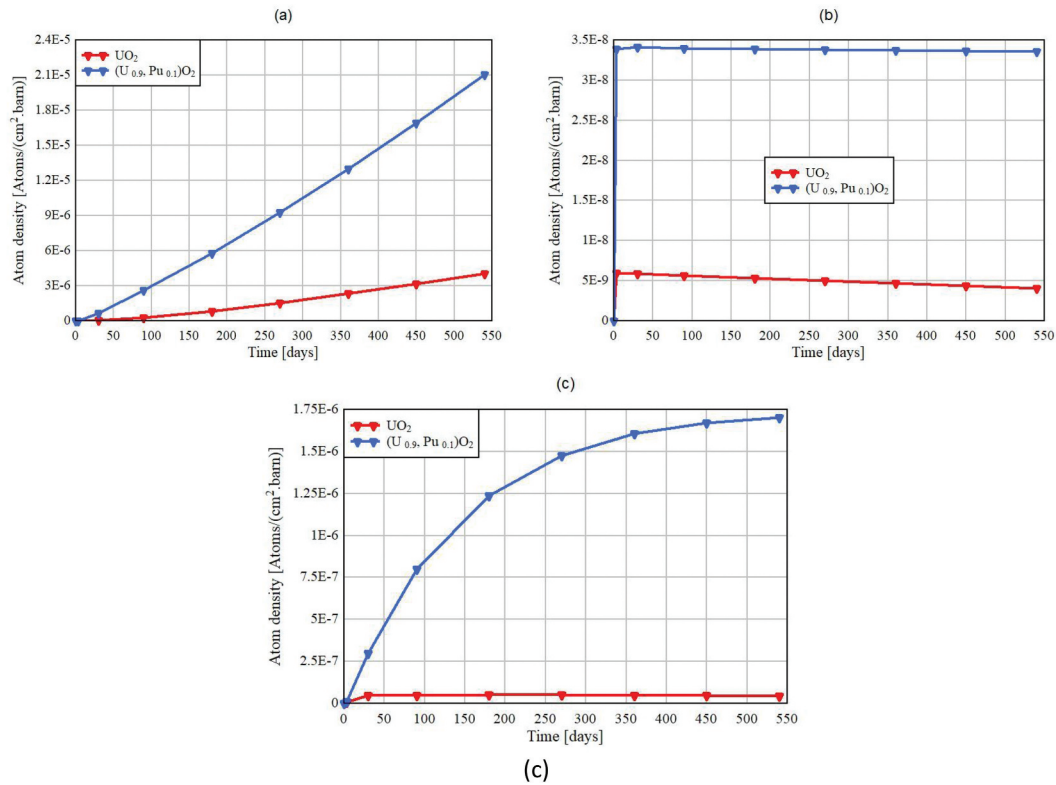


Fig. 3. Production of (a) minor actinides and nonactinides of (b) ^{135}Xe and (c) ^{149}Sm .

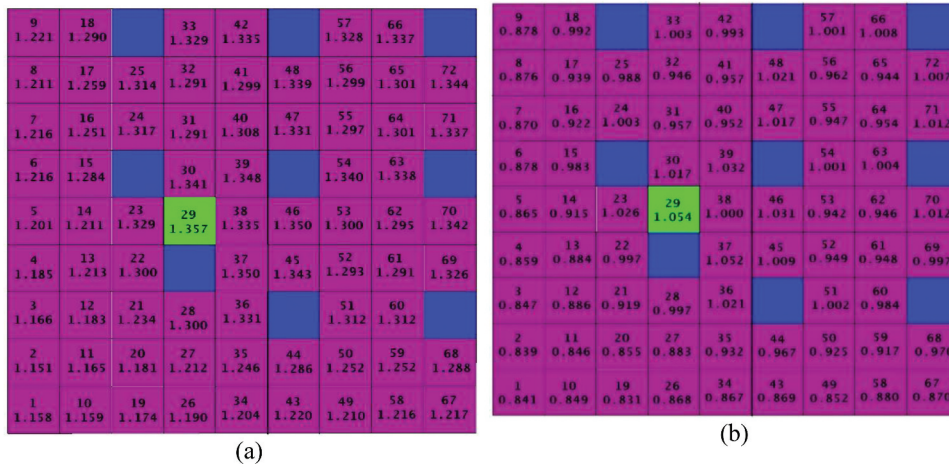


Fig. 4. Radial PPF profile for both (a) UO_2 and (b) $(\text{U}_{0.9}, \text{rgPu}_{0.1}) \text{O}_2$.

1.054. Despite generating similar thermal power levels, the MOX $[(\text{U}_{0.9}, \text{rgPu}_{0.1}) \text{O}_2]$ fuel exhibited a smaller maximum PPF compared to the UO_2 fuel.

This reduction in PPF in the case of the MOX fuel occurs because of two reasons. First, according to Eq. (12), for the same power, an increase in K_{inf} leads to a decrease in the normalization factor, resulting in a decrease in neutron flux, and hence, a lower PPF:

$$\phi \left[\frac{\text{neutrons}}{\text{cm}^2 \cdot \text{s}} \right] = \frac{\text{power}[\text{W}] \times \bar{v} \left[\frac{\text{neutrons}}{\text{fission}} \right] \times \phi_{F4} \left[\frac{1}{\text{cm}^2} \right]}{W_f \left[\frac{\text{W}}{\text{fission}} \right] \times K_{inf}[1]} \left[\frac{\text{neutrons}}{\text{cm}^2 \cdot \text{s}} \right], \quad (12)$$

where

ϕ = normalized flux

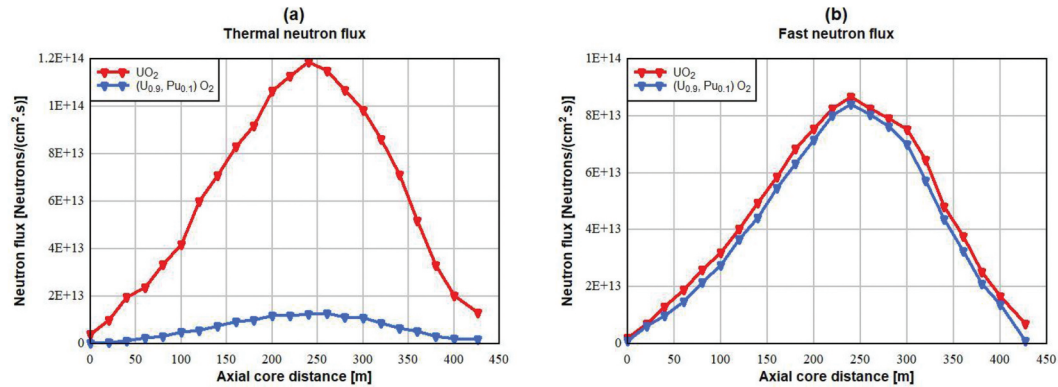


Fig. 5. Axial fluctuation for (a) thermal neutrons and (b) fast neutrons.

$\bar{\nu}$ = average neutron yield

ϕ_{F4} = neutron flux calculated by MCNPX

W_f = energy released per fission

K_{inf} = infinite multiplication factor.^[13]

Second, (U, Pu) O₂ fuel contains plutonium isotopes. Plutonium has a higher neutron capture cross section, especially for thermal neutrons. This means that plutonium is more likely to absorb thermal neutrons without causing fission. This neutron absorption by plutonium reduces the number of thermal neutrons, and hence, reduces the PPF. Equation (12) was used for calculating the normalized flux in MCNPX.

The variation of thermal neutron flux is illustrated in Fig. 5a, where the thermal neutron flux in the traditional fuel assembly surpasses that in the MOX assembly. The previously discussed two reasons for the reduction in PPF are the main causes for the large depression in thermal neutrons, specifically the second reason. Figure 5b displays the fast neutron flux, where the fast neutron flux in case of UO₂ is higher than that of (U, Pu) O₂. This is because of the normalized flux in the case of MOX is lower than that of UO₂. Also, the ²³⁹Pu and ²⁴¹Pu have higher absorption cross sections for fast neutrons.

III.B.3. Safety-Related Parameter Results

In this section, a safety comparison is conducted between the standard fuel material and the MOX fuel. Table II presents the calculation results for various safety-related parameters. The FTC for (U, Pu) O₂ differs from that of UO₂, primarily because it is mainly affected by the fissile material, specifically how resonance peaks impact neutron capture. In general, when the fuel temperature rises, the gap between nuclei increases, reducing the likelihood of fission, which in turn decreases reactivity.

TABLE II

Results of the Main Safety Parameters

Parameter	UO ₂	(U _{0.9} , rgPu _{0.1}) O ₂
FTC (pcm/°C)	-2.158	-1.924
MTC (pcm/°C)	-12.4	-10.2
β_{eff} (pcm)	775	449
Black CRW	-0.385	-0.19396
Gray CRW	-0.20746	-0.08832

However, resonance peaks behave differently. For ²³⁵U, these peaks generally occur at lower neutron energies, especially in the thermal neutron energy range (typically below 1 eV). In contrast, for ²³⁹Pu, resonance peaks tend to occur at higher neutron energies than ²³⁵U. Consequently, the presence of ²³⁹Pu decreases the capture cross section of thermal neutrons compared to ²³⁵U, increasing reactor reactivity due to the higher number of thermal neutrons.^[12,14] This, in turn, leads to an increase in the FTC for (U, Pu) O₂.

Regarding the MTC, the fuel temperature kept constant and the moderator temperature varied. An increase in moderator temperature results in a decrease in moderator density. This reduction in density diminishes the moderator's ability to slow down neutrons, which in turn reduces reactor reactivity. However, the presence of ²³⁹Pu in (U, Pu) O₂ decreases the capture cross section of thermal neutrons, leading to increased reactivity compared to UO₂. Eventually, UO₂ fuel typically possesses a FTC that is negative, enhancing the safety of thermal reactors. Conversely, the FTC for (U, Pu) O₂ fuel can fluctuate and may be either positive or slightly negative. So, it is crucial to thoroughly assess the specific behavior of (U, Pu) O₂ fuel when designing reactors and conducting safety analyses.

To evaluate the reactivity of the reactor during low-power operation, Eq. (13) is utilized to calculate the effective delayed neutron fraction β_{eff} :

$$\beta_{eff} = 1 - \left(\frac{K_P}{K} \right), \quad (13)$$

where K is the prompt and delayed neutrons effective multiplication factor, K_P is the effective prompt neutron multiplication factor, and β_{eff} mostly depends on the type of fissile material employed (^{235}U or ^{239}Pu). It is known that the β_{eff} for ^{239}Pu is much lower than that of ^{235}U ,^[15,16] so the use of PuO_2 decreased the delayed neutron fraction.

The investigation of control rod worth (CRW) is crucial to ensuring reactor safety during the design phase. In the AP-1000 nuclear power reactor, control rods consist of two different materials. The first material is an alloy known as Ag-In-Cd, which has a density of 10.159 g/cm^3 according to Ref. [11]. This alloy is utilized in both the rod cluster control assemblies (RCCAs) and the gray rod control assemblies (GRCAs). The radii of the (Ag-In-Cd) alloy within the RCCAs and GRCAs are 0.43307 cm and 0.20320 cm , respectively. The second material used is stainless steel 304, which possesses a density of 7.889 g/cm^3 . This stainless steel serves a dual purpose. First, it acts as a cladding material for the Pyrex rods, RCCA, and GRCA rods. Second, it functions as an absorbing material with a radius of 0.20320 cm within the GRCAs.

The RCCA rods, often referred to as “black rods,” play a crucial role in reactor shutdown procedures and are located within the 24 empty guide tubes. On the other hand, the GRCAs, known as “gray rods,” are employed to manage the reactor’s reactivity during its operational period. The distribution of GRCAs within the assemblies consists of 12 GRCAs of the Ag-In-Cd type and 12 GRCAs of the stainless steel 304 type, as specified in

Ref. [17]. The materials used in the control rods, as well as the fuel type, play a significant role in determining the reactivity changes upon control rod insertion.

The CRW was computed using Eq. (14),^[18,19] where k_{in} and k_{out} represent the multiplication factors of the assembly with and without the control rod, respectively,

$$\text{CRW} = K_{in} - K_{out}. \quad (14)$$

It was observed that the traditional control rod material (Cd-In-Ag) did not effectively shut down the reactor, as the K_{inf} value at full insertion of control rod remained greater than unity. Hence, alternative control materials were explored to identify the most suitable one for compatibility with the proposed fuel material ($\text{U}_{0.9}, \text{rgPu}_{0.1}$) O_2 .

Table III lists these materials, and it was found that B_4C exhibited the best performance among the proposed control materials. With the full insertion of B_4C , the K_{inf} value becomes less than unity (subcritical) with a high margin, meeting the critical condition required for reactor shutdown. The primary issue with utilizing B_4C arises from the generation of alpha particles as a result of boron interacting with neutrons. To address the challenges associated with alpha particle production from boron carbide control rods, reactor designers and operators typically employ various strategies, including

1. regularly inspecting and maintaining control rod materials to assess their condition and integrity
2. continuously monitoring and analyzing reactor performance to account for changes in the neutron economy caused by the use of boron carbide control rods
3. researching and developing advanced control rod materials that exhibit improved resistance to radiation damage and reduced alpha particle production.

It is worth emphasizing that although alpha particles can present difficulties in reactor design and operation, boron

TABLE III
CRW for the Various Control Materials to Be Used with ($\text{U}_{0.9}, \text{rgPu}_{0.1}$) O_2

Control Rod Type	K_{inf} at Hot Full Power Without Inserting the Control Rods Equals 1.20645			
	K_{inf} at Full Insertion	Black CRW	K_{inf} at Full Insertion	Gray CRW
Standard (Ag-In-Cd)	1.00043	−0.19396	1.10607	−0.08832
$\text{Dy}_2\text{O}_3 + \text{HfO}_2$	0.96968	−0.22471	1.09513	−0.09926
$\text{Dy}_2\text{O}_3 + \text{TiO}_2$	0.99696	−0.19743	1.11218	−0.08221
HfB_2	0.92888	−0.26872	1.07427	−0.12247
B_4C	0.89359	−0.3008	1.05774	−0.13879

carbide remains a widely utilized and effective control rod material due to its exceptional neutron absorption properties.

III.C. TH Results

The TH simulation was performed using COMSOL-Multiphysics with a two-dimensional (2-D) axisymmetric space dimension. Table IV provides correlations for the thermophysical and SS characteristics of the $(U_{0.9}, \text{rgPu}_{0.1}) \text{O}_2$,^[20–22] while for both UO_2 and ZIRLO,TM their correlations exist in the work we have already published.^[9] For design parameters and TH assumptions, the following published works have discussed them perfectly.^[9,10] The software’s library contains the thermo-physical characteristics of helium and water.^[6]

III.C.1. Fluid Dynamics Results

In this study, the Navier-Stokes PDEs were solved using COMSOL-Multiphysics employing the Reynolds-Average Navier-Stokes equation and the k-epsilon turbulence model.

The solution of the Navier-Stokes equations provides the velocity profile of the coolant and the decrease in coolant pressure along the axial direction. Figure 6a illustrates the coolant velocity profile at various heights, demonstrating how the profile changes from the entrance length at $Z = 0$ to $Z = 0.01H$ and then remains constant in the fully developed region. The 2-D velocity profile reveals the presence of the no-slip condition, where the coolant velocity is zero next to the static wall and increases as it moves away from the wall because of the reduced shear effects between water layers. Figure 6b displays the pressure drop along the axial core distance, with an overall pressure drop of 160.8152 kPa.

III.C.2. Heat Transfer Results

Figure 7 illustrates the temperature profiles along the fuel centerline, cladding inner surface, and coolant for UO_2 and $(U_{0.9}, \text{rgPu}_{0.1}) \text{O}_2$. The maximum temperature along the fuel centerline was found to be 1439.078 K for UO_2 and 1295.786 K for $(U_{0.9}, \text{rgPu}_{0.1}) \text{O}_2$, both of which are below their respective solidus temperatures. The maximum

TABLE IV
Thermophysical Properties of the $(U_{0.9}, \text{rgPu}_{0.1}) \text{O}_2$

$(U_{0.9}, \text{rgPu}_{0.1}) \text{O}_2$	
$K \left(\frac{\text{W}}{\text{mK}} \right)$	$\frac{1}{1.528 \times \sqrt{0.00931 - 0.1055 + 2.885 \times 10^{-4}T}} + 78.38 \times 10^{-12}T^3$
$C_p \left(\frac{\text{J}}{\text{kgK}} \right)$	$0.9 \times C_p(\text{UO}_2) + 0.1 \times C_p(\text{PuO}_2)$
$\text{Rho} \left(\frac{\text{kg}}{\text{m}^3} \right)$	$11080 \times [1 + 2.04 \times 10^{-5} \times (T-273) + 8.7 \times 10^{-9} \times (T-273)^2] - 1$
$E \text{ (Pa)}$	$9.21 \times 10^{10} - 4.05 \times 10^7 \times T$
ν	0.37
α	$9.828 \times 10^{-6} - 6.39 \times 10^{-10}T + 1.33 \times 10^{-12}T^2 - 1.757 \times 10^{-17}T^3$

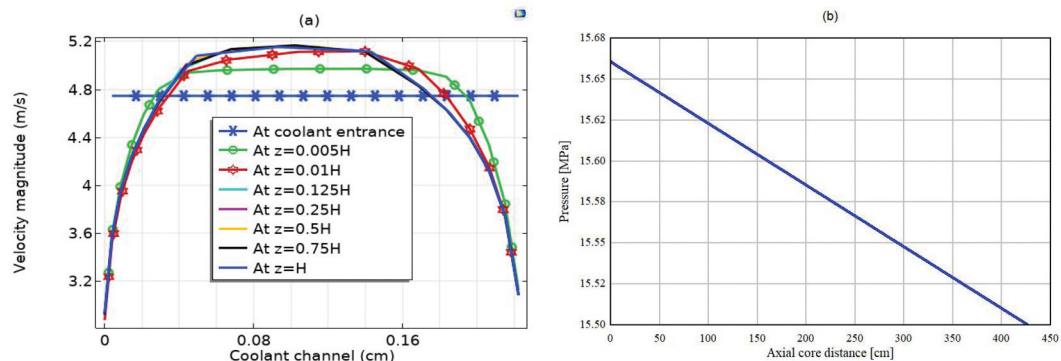


Fig. 6. Fluid mechanics outcomes for (a) coolant velocity profile and (b) coolant pressure drop.

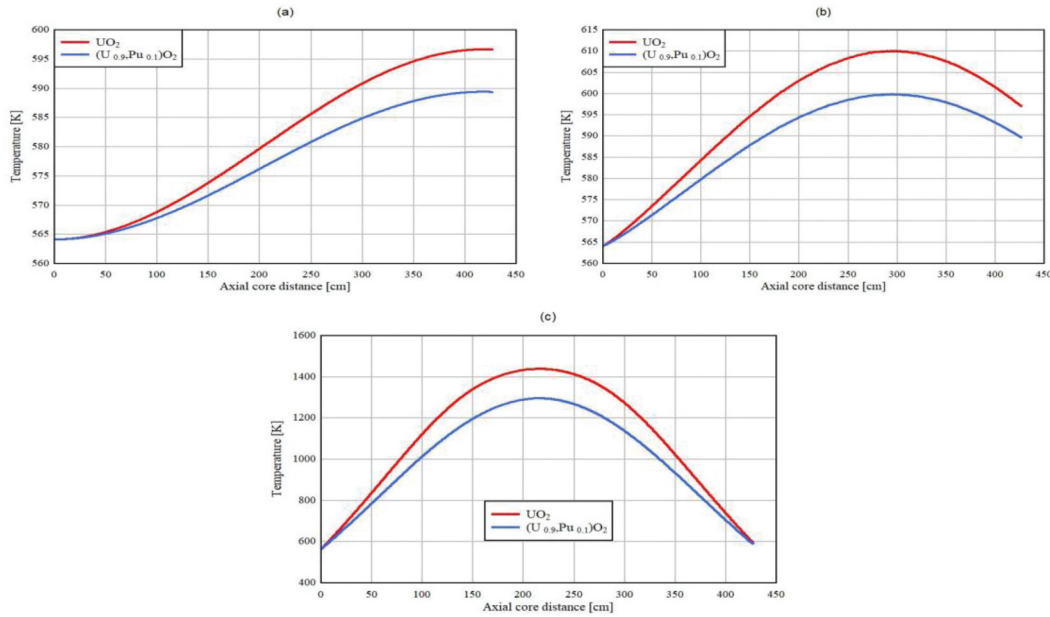


Fig. 7. Temperature profile for (a) coolant, (b) clad inner surface, and (c) fuel centerline.

temperature in the cladding occurred at 610.0459 K for UO₂ and 599.8312 K for (U_{0.9}, rgPu_{0.1}) O₂, which was below the melting point of ZIRLO at 2123 K. Based on the fuel and cladding temperature profiles, it can be concluded that the normal operational conditions ensure safe operation.

Figure 8 presents the three-dimensional (3-D) temperature profile of the channels with the maximum PPF. The critical heat flux q_{cri} was determined using the EPRI-1 correlation,^[1,3] and the DNBR was calculated by dividing q_{cri} by the q_{act} .

Figure 9 demonstrates that the minimum departure from nucleate boiling ratio (MDNBR) values for UO₂ and (U_{0.9}, rgPu_{0.1}) O₂ were 1.805 and 2.005, respectively, which indicates that the formation of nucleate boiling is not possible in the two investigated fuel assemblies. This

is confirmed by both MDNBR values exceeding the safety limit of 1.75 for PWRs.

III.D. Solid Mechanics Results

Figure 10a illustrates the highest von Mises stress experienced by the analyzed fuel materials, UO₂ and (U_{0.9}, rgPu_{0.1}) O₂, which were 131.6548 MPa and 116.7662 MPa, respectively. The fuel displacement was also simulated, with maximum displacements of 0.14259 mm for UO₂ and 0.11481 mm for (U_{0.9}, rgPu_{0.1}) O₂, as shown in Fig. 10b. The significant reduction in the structural strength parameters when using (U_{0.9}, rgPu_{0.1}) O₂ can be attributed to the substantial decrease in the maximum PPF, as depicted in Fig. 4,

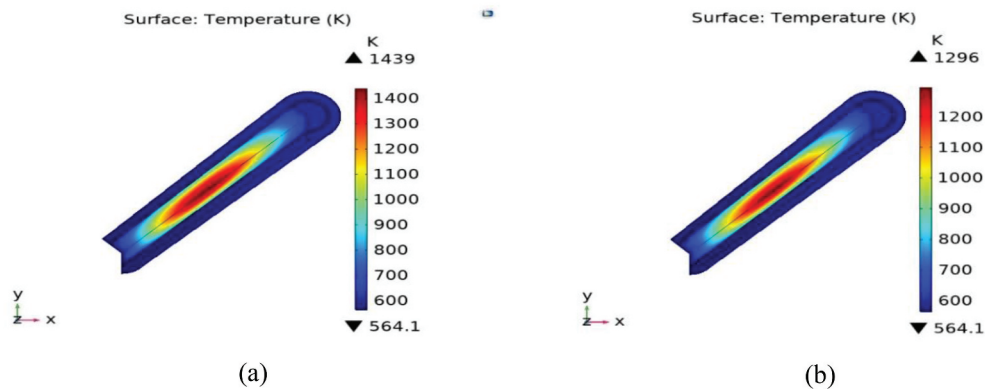


Fig. 8. Three-dimensional temperature distribution for (a) UO₂ and (b) (U_{0.9}, rgPu_{0.1}) O₂.

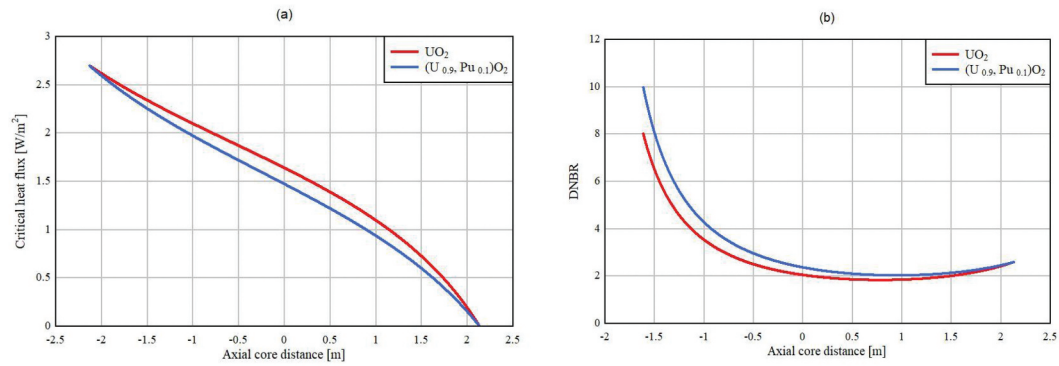


Fig. 9. Outcomes for DNBR simulation (a) q_{cri} and (b) DNBR.

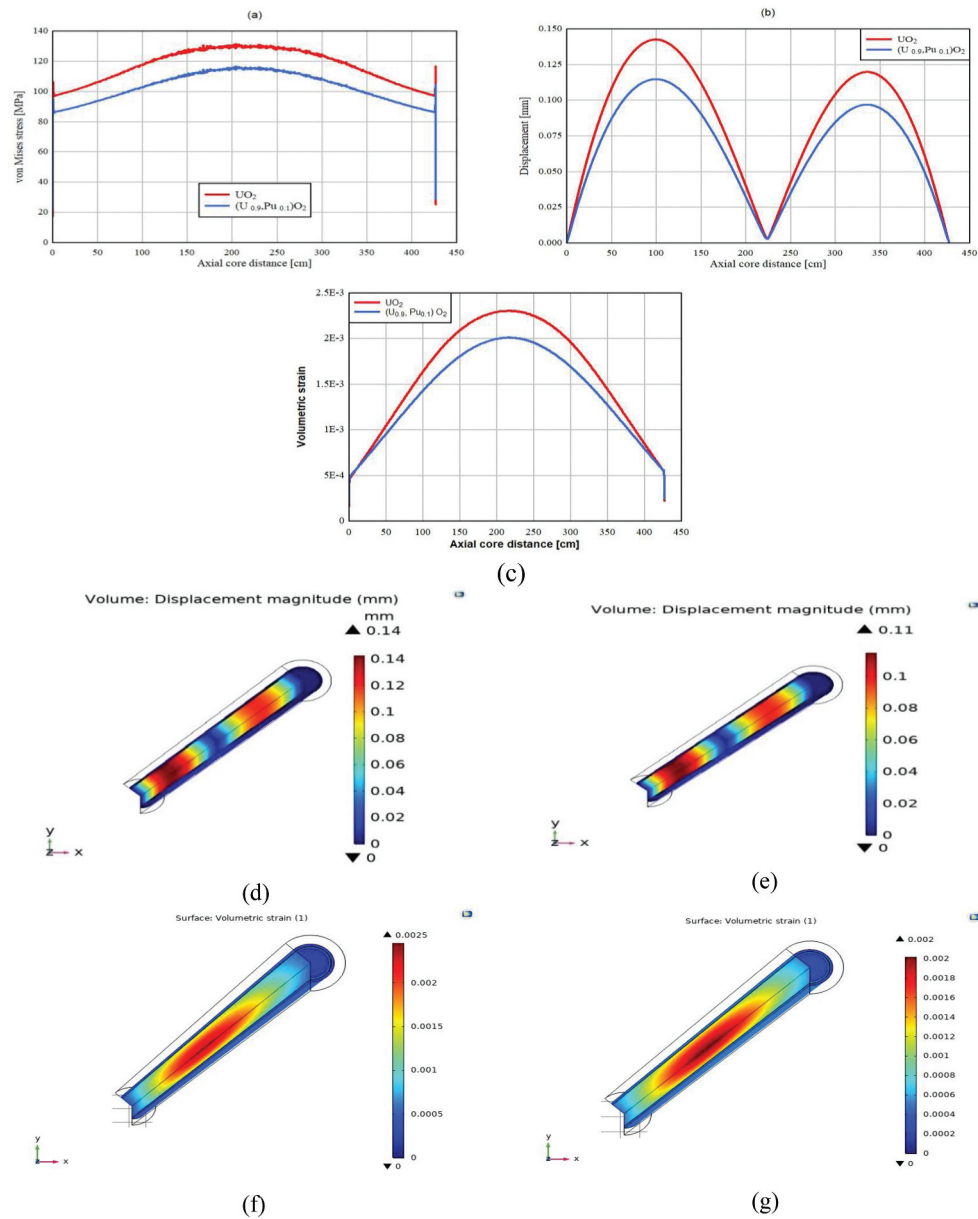


Fig. 10. Major SS outcomes for (a) stress, (b) displacement, (c) volumetric strain, (d) 3-D displacement for UO_2 , (e) 3-D displacement for $(U_{0.9}, rgPu_{0.1})O_2$, (f) 3-D volumetric strain for UO_2 , and (g) 3-D volumetric strain for $(U_{0.9}, rgPu_{0.1})O_2$.

TABLE V
Neutronic, TH, and SS Results Summary

Physics	Parameter	UO ₂	(U _{0.9} , rgPu _{0.1}) O ₂
Neutronic analysis	FCL (days)	480	960
	Maximum PPF	1.357	1.054
Heat transfer	Actinide inventory	Less	High
	Nonactinide inventory	Less	High
	Maximum fuel temperature (K)	1439.078; safety limit is 2800 K. ^[23]	1295.786; safety limit is 2774 K. ^[23]
	Maximum clad temperature (K)	610.0459	599.8312
		Safety limit is 2123 K. ^[23]	
	Maximum coolant temperature (K)	596.6515	589.3969
SS	MDNBR	1.805	2.005
		Safety limit is >1.75. ^[23]	
	Maximum stress (MPa)	131.6548	116.7662
		Safety limit is <200 MPa.	
	Maximum fuel displacement (mm)	0.14259	0.11481
		Safety limit is <0.08 mm.	
	Maximum volumetric strain	0.0025	0.002

rather than the material's intrinsic structural properties. The thermophysical and structural properties of UO₂ and (U_{0.9}, rgPu_{0.1}) O₂ are nearly identical, considering that the proportion of rgPuO₂ is small, suggesting that its influence on the material properties is minimal. Hence, it can be assumed that both materials exhibit similar structural properties.

Despite the contact between the fuel and cladding surfaces, uranium and zirconium will not chemically interact since the maximum cladding temperature remains below the interaction point of 1132 K. This contact phenomenon is advantageous for heat transfer, as it reduces the thermal resistance and enhances thermal conductivity, thereby lowering the fuel temperature.

Figure 10c presents the volumetric strain along the centerline of the fuel materials, where the maximum volumetric strain values occur are 0.0025 and 0.002 for both UO₂ and (U_{0.9}, rgPu_{0.1}) O₂, respectively. Figures 10d and 10e present the 3-D results of the fuel and cladding displacements in UO₂ and (U_{0.9}, rgPu_{0.1}) O₂, respectively. The 3-D results for the volumetric strain for both UO₂ and (U_{0.9}, rgPu_{0.1}) O₂ in Figs. 10f and 10g, respectively. The summarized results of the neutronic, TH, and structural analyses for both UO₂ and (U_{0.9}, rgPu_{0.1}) O₂ are provided in Table V. According to the von-Mises yield criterion, if the von-Mises stress acting

on material due to tension is equal or greater than the yield limit of the same material, the material will yield.

IV. CONCLUSION

This study conducted a comprehensive analysis to assess the potential advantages of employing (U_{0.9}, rgPu_{0.1}) O₂. From a neutronic perspective, (U_{0.9}, rgPu_{0.1}) O₂ exhibited a higher fuel conversion ratio (FCL) compared to standard fuel, approximately three times greater. However, the use of (U_{0.9}, rgPu_{0.1}) O₂ resulted in increased production of both actinide and nonactinide nuclides.

In terms of reactivity control and reactor shutdown, B₄C proved to be the most effective control material when used in conjunction with (U_{0.9}, rgPu_{0.1}) O₂, surpassing the performance of the typical control material (Ag-In-Cd) employed in the AP-1000 reactor when used with the MOX fuel. The utilization of (U_{0.9}, rgPu_{0.1}) O₂ as fuel led to a significant reduction in the maximum PPF, resulting in improved TH and SS performance.

Nomenclature

β_{eff} = effective prompt multiplication factor

K_{inf} = infinite multiplication factor

Acronyms

2-D	= two dimensional
3-D	= three dimensional
ATF	= accident-tolerant fuel
CRW	= control rod worth
DCD	= design control document
DNBR	= departure from nucleate boiling ratio
FCL	= fuel cycle length
FEM	= finite element method
FEPDs	= full effective power days
FTC	= fuel temperature coefficient
GRCA	= gray rod control assemblies
MDNBR	= minimum departure from nuclear boiling ratio
MOX	= mixed oxide
MTC	= moderator temperature coefficient
PDE	= partial differential equation
PPF	= power peaking factor
PWR	= pressurized water reactor
RCCA	= rod cluster control assemblies
rgPu	= reactor-grade plutonium
SS	= solid structure
TH	= thermal hydraulic

Disclosure Statement

No potential conflict of interest was reported by the authors.

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